

Proof Attempt: ~~Let~~ $(x \circ y) \circ z \subseteq x \circ (y \circ z)$ is always true.

For the other side: To show $x \circ (y \circ z) \subseteq (x \circ y) \circ z$, one must show

$$R_1 \circ (S_1 \circ T_1) \subseteq (R_1 \circ S_1) \circ T_1 \text{ and } R_2 \circ (S_2 \circ T_2) \subseteq (R_2 \circ S_2) \circ T_2$$

$$\text{Let } (p, T) \in R_1 \circ (S_1 \circ T_1)$$

$$\exists U \ (p, U) \in R_1 \text{ and } (U, T) \in (S_1 \circ T_1) \ \forall u \in U$$

$$\forall u \in U, \exists T_u \text{ s.t. } (u, T_u) \in S_1 \text{ and } (t', T) \in T_1 \ \forall t' \in T_u.$$

~~Then by downward monotonicity~~

$$\text{Let } U' = \cap T_u$$

Then by downward monotonicity $(u, \cap T_u) \in S_1 \ \forall u \in U$.

$$\therefore (p, \cap T_u) \in R_1 \circ S_1$$

If nT_u is non-empty, no-problem with the below:

$$\forall t' \in nT_u, (t', T) \in T_1.$$

$$\therefore (p, T) \in (R \circ S_1) \circ T_1.$$

$$\begin{aligned} t' \in nT_u &\Rightarrow (t', T) \in T_1 \\ t' \in nT_u &\Rightarrow (t', T) \in T_1 \end{aligned}$$

If nT_u is empty,

then $t' \in nT_u \Rightarrow (t', T) \in T_1$ is vacuously true.

$$\text{So, } (p, T) \in (R_1 \circ S_1) \circ T_1.$$

Some Observations:

① R empty iff $R \circ S$ empty.

Proof: Suppose R empty.

Claim: $R \circ S$ empty.

Otherwise, let $(p, T) \in R \circ S$

$\therefore (p, U) \in R$ f.s. U

but R is empty!

Conversely, let $R \circ S$ be empty.

Claim: R empty.

Otherwise let $(p, T) \in R$.

$\therefore (p, \emptyset) \in R$ since $\emptyset \subseteq T$ and by downward monotonicity

$\forall x \in \emptyset$, ~~$(x, T) \in S$~~ $(x, T) \in S$ is vacuously true.

$\therefore (p, T) \in S$

contradiction!

② If $(p, T) \in R$, then $(p, X) \in R \circ S$ for all sets X .

Proof: Let $(p, T) \in R$.

$\therefore (p, \phi) \in R$ since $\phi \subseteq T$ and by downward monotonicity.

$\forall v \in \phi, (v, X) \in S$ is vacuously true.

$\therefore (p, X) \in S$ where X is any set.

③ Combining the above two facts,

$R \subseteq R \circ S$ for all R, S .

$(p, T) \in R \circ S$.